Analysis of a Multilingual Language-Model Project

Telugu (Model H) and Assamese (Model L)

Individual Project Analysis

September 2026

Abstract

This report records the project as three connected phases: corpus construction and tokenizer selection, Transformer pretraining and language-model evaluation, and reasoning fine-tuning with behavioural and attention analysis. The two models use matched architecture and capacity, allowing their language-model behaviour to be compared while keeping the differences in corpus composition visible. The final phase is treated as an experimental progression rather than only a final result: five fine-tuning configurations were tested, and the failures of the earlier configurations explain the changes introduced later. In addition to accuracy, the analysis examines seen versus held-out entities, reasoning depth, attention entropy, attention distance and attention paid to entity tokens. The aim is to describe what the measurements establish, what they suggest, and where the evidence is not sufficient for a stronger claim.

1 Project Overview

The central idea of the project was to train small Transformer language models independently for Telugu and Assamese and then investigate whether the pretrained representations could be adapted to a controlled reasoning benchmark. The two models were intentionally kept at the same capacity: 12 Transformer layers, 6 attention heads, model dimension 384, context length 512 and an 8K vocabulary, for a total of 24,562,944 parameters. This matched setup is useful because it removes architecture size as an obvious explanation for differences between the languages.

The comparison is not a perfectly controlled language experiment, however. The corpora themselves are different. Telugu contains more training documents and is strongly dominated by Sangraha, while Assamese contains fewer documents but a large Archive/OCR component. Document-length distributions also differ. These properties affect the sequence distribution presented to the model, so later differences in perplexity are interpreted together with the corpus analysis rather than as evidence that one language is intrinsically easier.

The work was divided into three phases. Phase 1 prepared the data and selected a tokenizer. Phase 2 trained and evaluated the language models. Phase 3 used synthetic reasoning examples to adapt the models and then examined both the output behaviour and changes in attention. This structure is important because each phase supplies a baseline for the next one.

2 PHASE 1 – Data Collection, Corpus Construction and Tokenization

2.1 1.1 Building usable corpora

The first phase involved more than collecting text. The sources had to be filtered and normalized so that the resulting corpus could be used consistently for language-model training. The pipeline used Sangraha, Wikipedia, Wikisource, Internet Archive OCR and Telugu textbook material. Script and quality filtering, exact deduplication, near-deduplication and source labelling were applied before deterministic train/validation/test splitting.

The final Telugu training split contains 1,015,350 documents and 647,207,005 tokens. Assamese contains 392,097 documents and 559,359,252 tokens. Thus the Assamese corpus has fewer documents but a comparable token count. This difference is a useful reminder that token count alone does not describe the data distribution.

The source-composition plot shows that Sangraha contributes about 504.7M Telugu training tokens. The remaining reported sources contribute approximately 63.6M Wikipedia tokens, 54.9M Archive tokens, 15.0M Wikisource tokens and 9.0M textbook tokens. Sangraha therefore dominates the aggregate training distribution. The split-size plot answers a different question: it shows how the data were allocated to evaluation splits and how the manual-collection component entered those splits. Both views are needed because source composition and split composition are not interchangeable.

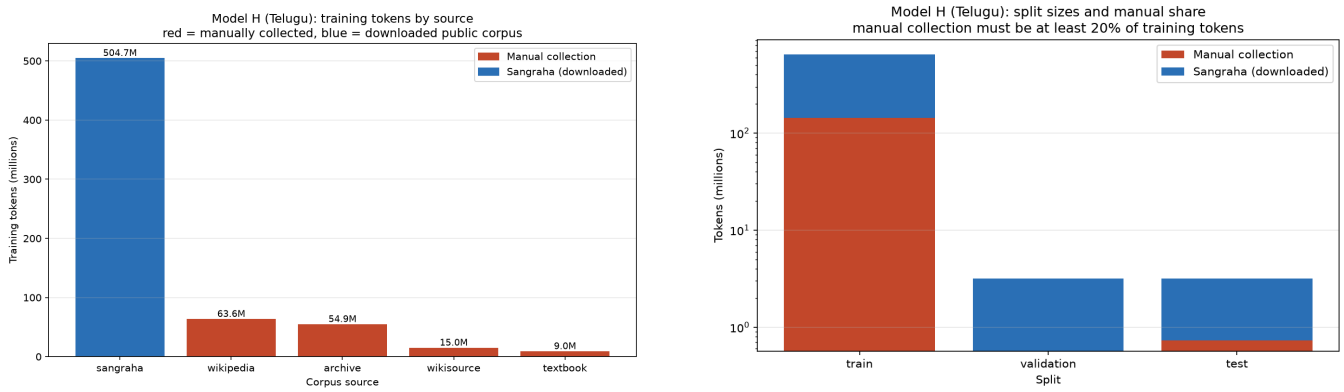


Figure 1: Phase 1 corpus composition for Model H (Telugu). The left panel shows training tokens by source; the right panel shows the train, validation and test allocation with manual and downloaded components.

2.2 1.2 Document length and source structure

The Assamese corpus contains a substantial Archive/OCR component. The document-length analysis shows why this matters: different sources occupy different length ranges rather than forming one homogeneous distribution.

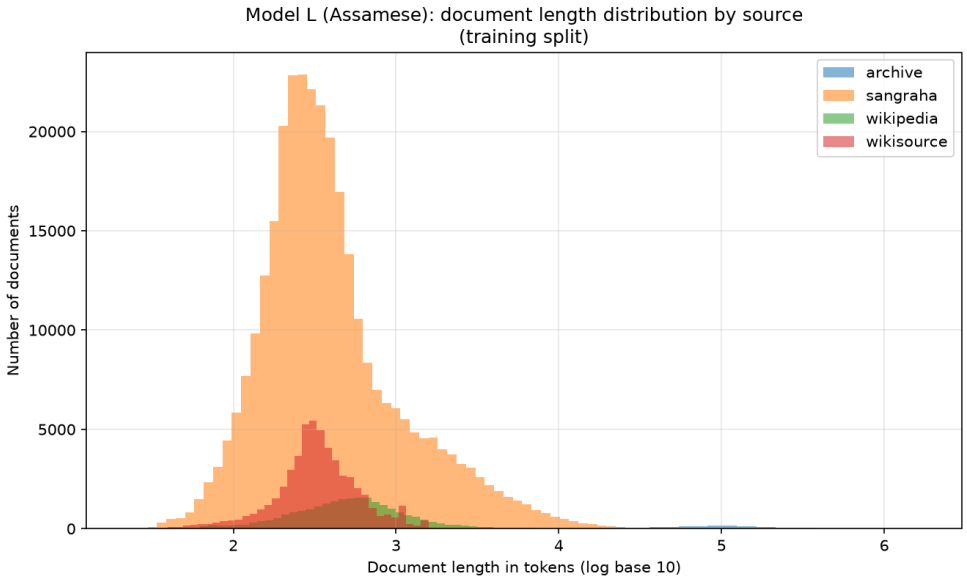


Figure 2: Model L (Assamese): document-length distribution by source in the training split. The x-axis is $\log_{10}$ document length in tokens.

In Figure 2, Sangraha forms the largest mass of documents around moderate lengths, whereas Archive extends much further into the long-document region. Wikisource and Wikipedia occupy their own ranges as well. This is relevant to pretraining because the model is exposed to different amounts of context and different source styles. The later difficulty of Archive should therefore be viewed as a source-level observation. It is reasonable to suspect that OCR and document structure contribute, but the current experiment does not independently vary OCR quality, source and length, so it cannot identify one of these variables as the sole cause.

2.3 1.3 Tokenizer vocabulary sweep

The tokenizer vocabulary was selected independently for each language rather than assuming that the same vocabulary size would be appropriate for both models. SentencePiece Unigram candidates at 6K, 8K, 10K and 12K were trained on a 10% sample of the respective training split and evaluated on held-out validation text. Fertility, characters per token, unknown-token rate and byte-fallback rate were considered. The results for Assamese are shown below because Model L made its own vocabulary decision from its own measurements.

For Assamese, increasing the vocabulary size steadily improves subword compression. The measured values are 2.4939 characters per token and 2.1034 fertility at 6K; 2.6117 characters per token and 2.0085 fertility at 8K; 2.6980 characters per token and 1.9443 fertility at 10K; and 2.7646 characters per token and 1.8974 fertility at 12K. Thus, the larger vocabularies do reduce the number of token pieces required to represent the text, but the improvement becomes progressively smaller.

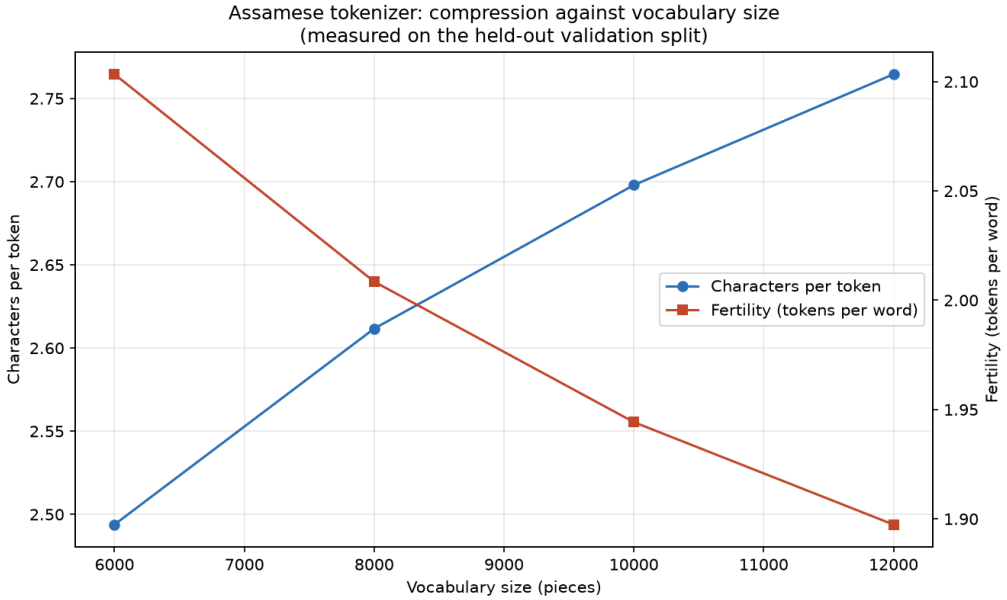


Figure 3: Assamese (Model L): tokenizer compression against vocabulary size on the held-out validation split.

Assamese vocabulary selection

The 8K vocabulary was not selected because it produced the lowest fertility. In fact, both 10K and 12K have lower fertility than 8K. The decision was made together with the fixed approximately 25M parameter budget used in Phase 2. With $d_{model} = 384$, the token embedding alone requires vocab $\times$ 384 parameters. For Model L, the sweep corresponds to 2.304M parameters at 6K, 3.072M at 8K, 3.840M at 10K and 4.608M at 12K. The larger embedding therefore consumes parameters that could otherwise be used by the Transformer layers.

The capacity trade-off is particularly important because the architecture is fixed at 12 layers, 6 heads and $d_{model} = 384$. At 6K and 8K, the model can retain all twelve Transformer layers. At 10K and 12K, the additional embedding cost reduces the affordable depth to eleven layers within the same parameter budget. In other words, the vocabulary is effectively bought out of Transformer capacity. The 8K choice therefore provides a useful compromise: it improves substantially over 6K while preserving the full twelve-layer architecture, whereas moving to 10K or 12K gains further compression at the cost of an entire Transformer block.

This same principle was applied independently to both languages. Telugu and Assamese did not share tokenizer selection measurements; each language was evaluated on its own corpus and the final architecture used 8K for both. For Assamese, the measured sweep therefore supports 8K as the point where the compression benefit remains useful without sacrificing a Transformer layer under the approximately 25M parameter constraint.

2.4 1.4 Phase 1 interpretation

The main result of Phase 1 is that the two models do not start from equivalent statistical problems. Telugu has a strong Sangraha contribution, while Assamese has a large Archive/OCR component and fewer, longer documents. The tokenizer experiment also shows that preprocessing choices directly affect the small-model capacity budget. These facts become necessary context when interpreting Phase 2.

3 PHASE 2 – Transformer Pretraining and Language-Model Evaluation

3.1 2.1 Matched architecture and optimization

Both models use 24,562,944 parameters, 12 layers, 6 heads, $d_{model} = 384$, context 512 and vocabulary 8K. Pretraining used AdamW, a peak learning rate of 6×10^{-4}, linear warmup followed by cosine decay, weight decay 0.1, gradient clipping at 1.0 and FP16 computation.

The schedule rises quickly to the peak learning rate and then decays smoothly. Both models follow the same schedule shape. Therefore, the different validation trajectories cannot be explained by a different learning-rate schedule.

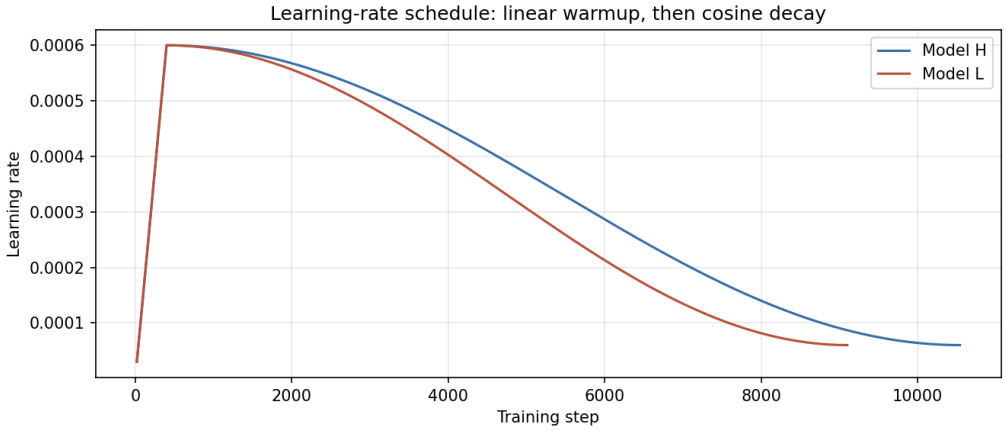


Figure 4: Pretraining learning-rate schedule: linear warmup followed by cosine decay for Models H and L.

3.2 2.2 Perplexity during pretraining

Perplexity is obtained from token-level cross-entropy and summarizes predictive uncertainty. Lower perplexity means that the model assigns higher probability to the observed evaluation sequence.

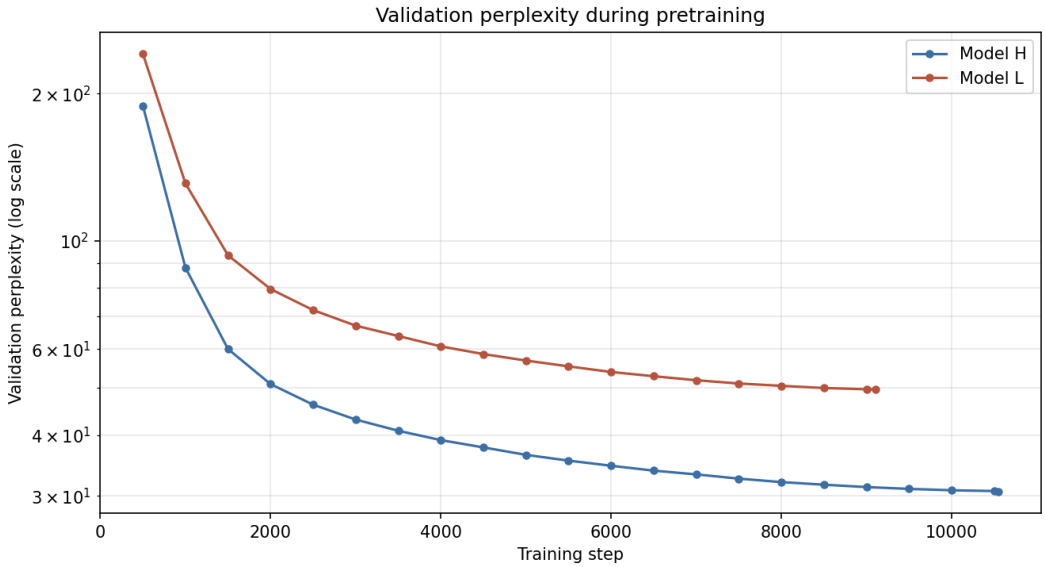


Figure 5: Validation perplexity during pretraining. The y-axis uses a logarithmic scale.

The curves fall rapidly during the early training steps and then improve more slowly. Model H remains below Model L throughout the displayed trajectory. The flattening near the end indicates diminishing improvement under the same training setup.

Metric	Model H (Telugu)	Model L (Assamese)
Test cross-entropy	3.4891	3.8859
Perplexity	32.7566	48.7110
Bits/token	5.0337	5.6062
Bits/byte	0.5369	0.6509

Table 1: Phase 2 final language-model evaluation on the complete test split.

Model H consequently has lower reported cross-entropy, perplexity, bits/token and bits/byte. Since architecture and capacity are matched, the result is consistent with a difference arising from the training distributions and data quality/structure. It should not be interpreted as an intrinsic ranking of the languages.

3.3 2.3 Source-wise evaluation and generation

Aggregate perplexity can hide source-specific behaviour. The project therefore evaluated sources separately. Archive was difficult for both models, and the Assamese corpus contains a large Archive component. This provides evidence

consistent with a corpus-composition effect. It does not establish that OCR quality alone caused the gap because source, genre, length and text quality are entangled.

Generation was evaluated with BLEU-4, chrF2, ROUGE-L, Distinct-1, Distinct-2 and 4-gram repetition at different temperatures. These measures were treated as secondary evidence. Lower temperatures generally increase reference overlap, whereas higher temperatures increase diversity. A single generation metric therefore cannot be used as the central language-model result.

3.4 2.4 Attention baseline before reasoning adaptation

Attention was analyzed across all twelve layers using normalized entropy and mean attention distance. Entropy measures how concentrated the attention distribution is; distance measures how far the attended key positions are from the query position. They answer different questions.

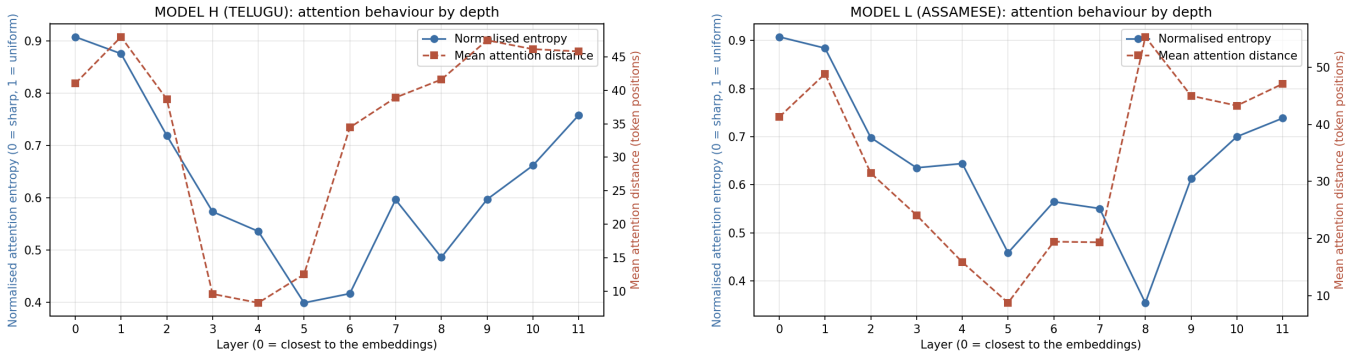


Figure 6: Pretraining attention baseline: Model H (left) and Model L (right). Both show strong depth-dependent changes in entropy and attention distance.

For Model H, normalized entropy begins high near the embedding layer, falls substantially through the middle layers and rises again toward the top. Mean attention distance is relatively short around the middle and larger in upper layers. Model L shows a comparable broad structure, with entropy reaching a low around layer 8 and attention distance becoming large around the same upper-middle region. The reported median normalized entropy is 0.6606 for Telugu and 0.6706 for Assamese. These values establish the reference against which the Phase 3 changes are measured.

4 PHASE 3 – Reasoning Fine-tuning and Analysis

4.1 3.1 Reasoning benchmark and evaluation design

The benchmark contains numeric pair, relation and extreme tasks together with chain pair, extreme and yes/no tasks. Examples require one, two or three inference hops. The reported chance exact-match level is 0.410.

The evaluation separates familiar and held-out entities. In `test_seen`, entity names were familiar from training but prompts were unseen. In the held-out test, the entities themselves were not present during training. This is an important control: it reduces the possibility that high performance is explained entirely by memorizing a fixed set of names.

4.2 3.2 Five-run fine-tuning progression

The five runs are retained because each exposed a different failure mode. The project did not simply choose the last checkpoint; the configuration was changed in response to what the previous experiment showed.

Run 3 demonstrates the cost of aggressive adaptation particularly clearly. In the separate 600-window language-model retention evaluation used in Phase 3, perplexity increased from the pretrained baselines of 13.79 to 6709.43 for Telugu and from 61.94 to 868303.09 for Assamese. These pretrained baselines belong to the Phase 3 retention evaluation, not to the complete Phase 2 test-set evaluation reported in Table 1. The change is not ordinary overfitting in the sense of a small validation gap; the original language-model distribution was severely displaced.

Run 5 used 8,000 examples, 6,000 steps, batch size 16, peak learning rate 2×10^{-5}, 20% warmup, weight decay 0.03 and curriculum sampling. The change was therefore not simply "train less." It combined smaller updates with a controlled warmup and a different sampling progression.

Run	Main change	Observation
1	Small pool	Memorised training names; name-answer EM was 0.
2	Larger pool	More/less/equal behaviour improved, but name selection remained near chance.
3	Worked answers, aggressive setup	Reasoning improved strongly but language-model performance collapsed.
4a	Gentler LR and fewer steps	LM damage reduced, but reasoning remained weak.
4b	Small-data variant	Weak-reasoning pattern persisted.
5	Low LR, warmup and curriculum	Better measured balance between reasoning and LM retention.

Table 2: Fine-tuning progression.

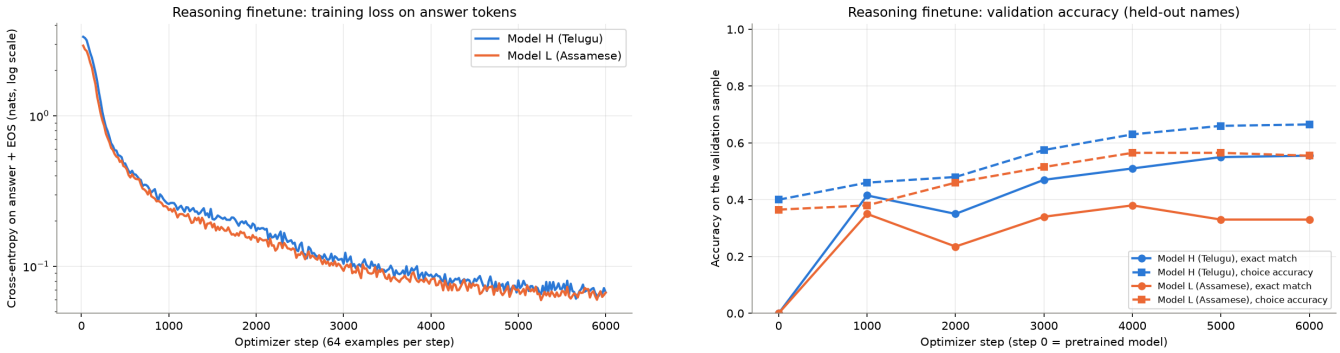


Figure 7: Run 5 training loss and held-out-name validation accuracy. The loss decreases strongly while validation performance improves with a non-monotonic trajectory.

4.3 3.3 Training loss and validation behaviour

The Run 5 loss decreases rapidly at the beginning and then more slowly, with minibatch fluctuations. This confirms that the model is fitting the reasoning answer tokens. It does not, by itself, demonstrate generalization because training loss can decrease through memorization.

The validation curve therefore matters. For Model H, held-out exact match rises from zero at initialization to about 0.42 near 1,000 steps, dips around 2,000 steps and then increases to approximately 0.55 near the end. Model L follows a weaker pattern, reaching roughly 0.38 around 4,000 steps and about 0.33 at the final displayed points. Choice accuracy is higher than exact match for both models, showing that the model can sometimes identify the correct choice without reproducing the exact expected answer string.

4.4 3.4 Seen versus held-out entities

The final test results provide a more direct measure of generalization. Seen-name exact match is 0.706 for Telugu and 0.656 for Assamese. Held-out exact match is 0.493 and 0.388. The corresponding drops are 0.213 and 0.268.

This gap is an important part of the result. The models learn useful behaviour on the synthetic benchmark, but the behaviour transfers less effectively when the entity names are replaced by entities never encountered during training. Therefore, high seen-name performance should not be interpreted as pure evidence of an abstract reasoning procedure.

4.5 3.5 Task-wise reasoning performance

The task-wise plot shows that adaptation is uneven. Numeric relation is particularly strong, while chain tasks are more difficult. The reported held-out numeric-relation exact match is approximately 0.915 for Telugu and 0.850 for Assamese. In contrast, chain-extreme exact match is only about 0.140 and 0.110. The model has therefore learned some relational operations much more reliably than multi-step composition.

This distinction is useful because an overall accuracy number could hide the actual failure mode. A model can learn the format of direct comparison questions while still failing when the answer requires maintaining an intermediate relation across several steps.

4.6 3.6 Reasoning depth

The hop-wise curve gives a consistent degradation pattern. Model H seen-name exact match falls from about 0.90 at one hop to 0.53 at two hops and 0.46 at three hops. Its held-out score falls from about 0.75 to 0.27 and then 0.15.

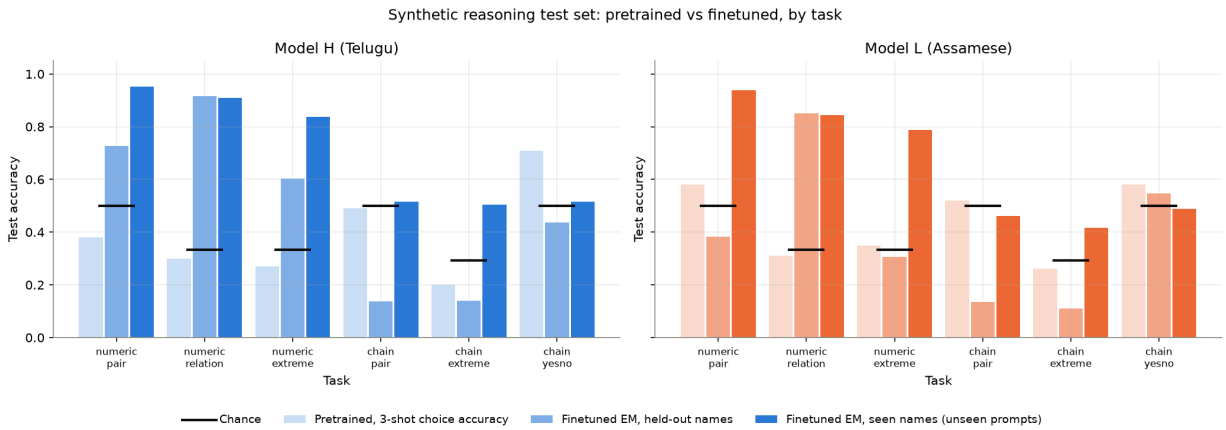


Figure 8: Synthetic reasoning test results by task for Models H and L. The plot compares pretrained choice accuracy with fine-tuned exact match for seen and held-out entities.

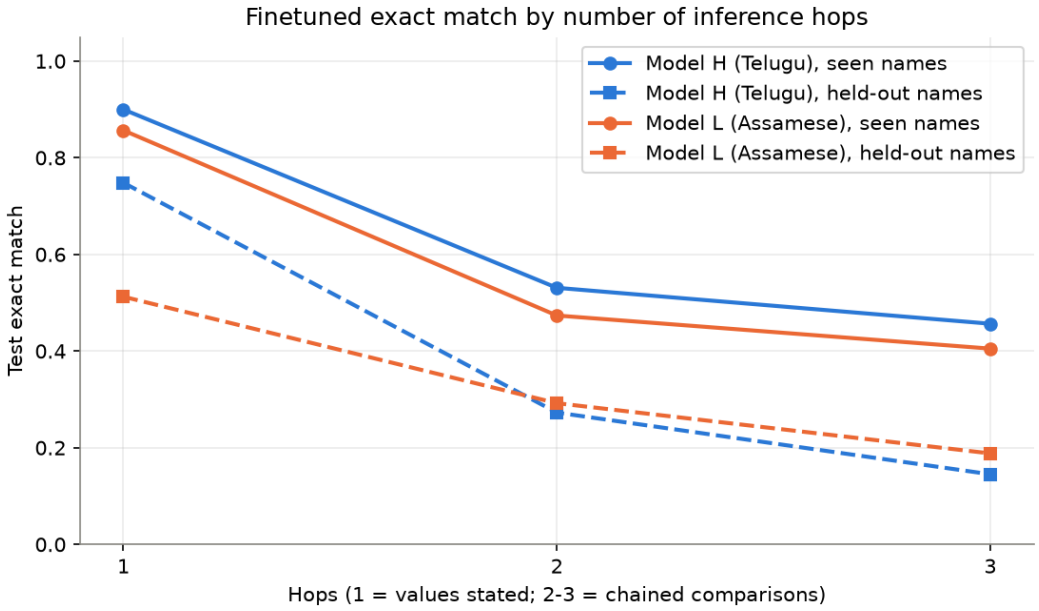


Figure 9: Fine-tuned exact match by inference depth. Solid lines are seen entities and dashed lines are held-out entities.

Model L shows a similar decrease: seen-name performance falls from about 0.86 to 0.48 and 0.40, while held-out performance falls from about 0.51 to 0.29 and 0.19.

The simplest interpretation is a composition bottleneck. One-hop examples mainly require a direct comparison. Two- and three-hop examples require the model to preserve an intermediate result and apply another relation. Prediction inspection also identified relational-direction mistakes, especially around "less than" statements. This is stronger evidence than merely observing a lower score: it identifies a concrete class of errors that can accumulate across a chain.

4.7 3.7 Attention to entity tokens after fine-tuning

The behavioural results show that fine-tuning changes what the model can do. The next question is whether the internal attention distribution changes in a way that is consistent with the new task.

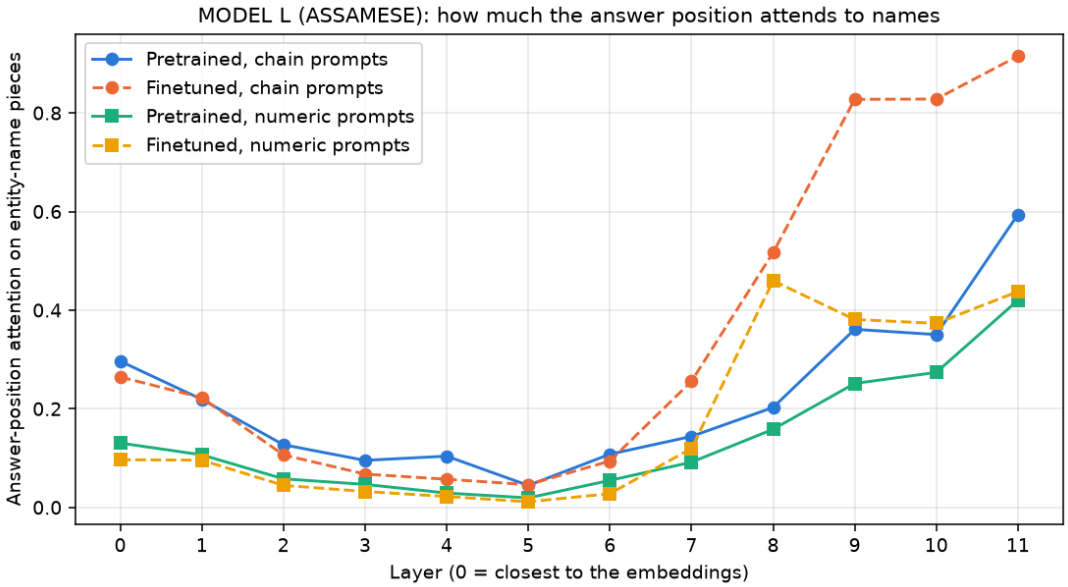


Figure 10: Model L: attention from the answer position to entity-name pieces across layers for pretrained and fine-tuned reasoning prompts.

For the Assamese model, fine-tuned chain prompts show a strong rise in answer-position attention to entity names in the upper layers. The fine-tuned chain curve reaches about 0.52 at layer 8, rises above 0.8 at layers 9 and 10, and reaches roughly 0.91 at layer 11. The corresponding pretrained chain curve reaches only about 0.59 at layer 11. Numeric prompts show a smaller but still visible late-layer increase.

The aggregate measurements for Telugu show similar changes: layer 9 increases from 0.264 to 0.582 and layer 10 from 0.449 to 0.650. For Assamese, corresponding changes include layer 9 from 0.306 to 0.604 and layer 11 from 0.507 to 0.677. The consistent direction is stronger late-layer interaction with the entity representations used to determine the answer.

This is evidence of internal adaptation, not proof of causal reasoning. Attention weights alone cannot establish that a particular head or layer performs the reasoning operation.

4.8 3.8 Attention entropy and selectivity

Normalized entropy is lower when attention is concentrated on fewer positions. Figure 11 shows that fine-tuning reduces entropy across many middle and upper layers. At layer 5, the aggregate value changes from about 0.56 to 0.45; at layer 6 it changes from about 0.57 to 0.47. The reduction is not uniform across every layer, but the middle network becomes noticeably more selective.

The per-head heatmap shows that the effect is structured. At layer 5, head 0 changes by approximately -0.22 and head 1 by -0.18. At layer 6, head 5 changes by about -0.24. Other heads are close to zero or can even become slightly more diffuse. Therefore, fine-tuning does not simply sharpen all heads equally; particular heads appear to be modified more strongly.

4.9 3.9 Attention distance and redistribution of range

Mean attention distance measures a different property from entropy. The fine-tuned model does not become uniformly more local or uniformly more long-range. In the upper layers, the large pretrained peak is reduced. At layer 8, for

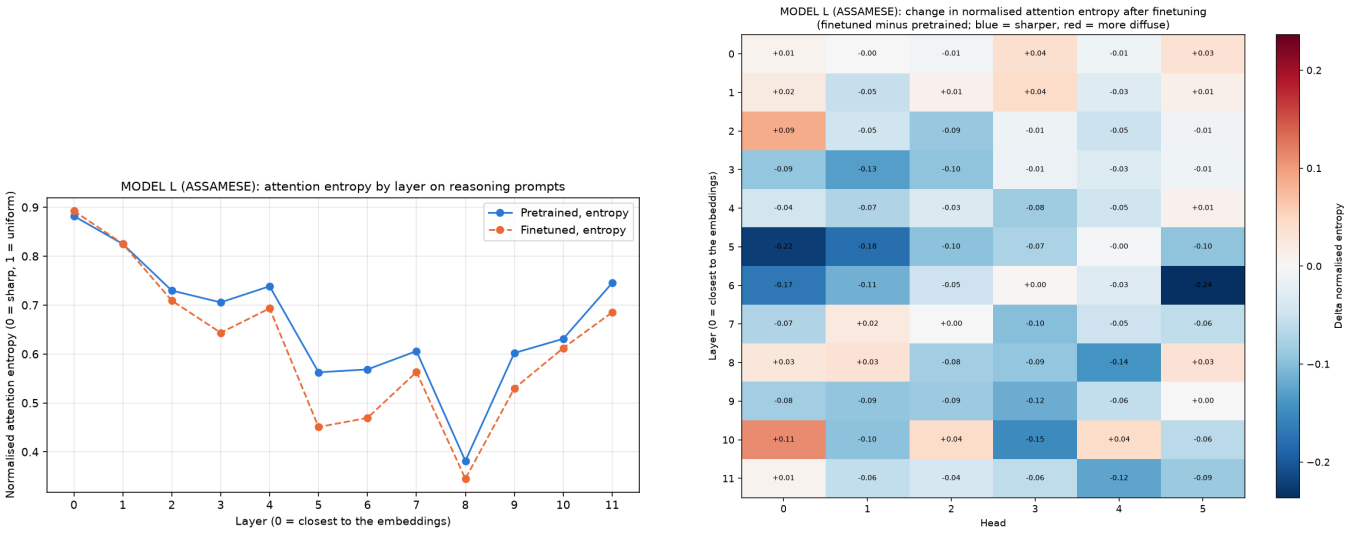


Figure 11: Model L attention entropy before/after fine-tuning and the per-head entropy change. Negative delta values indicate sharper attention.

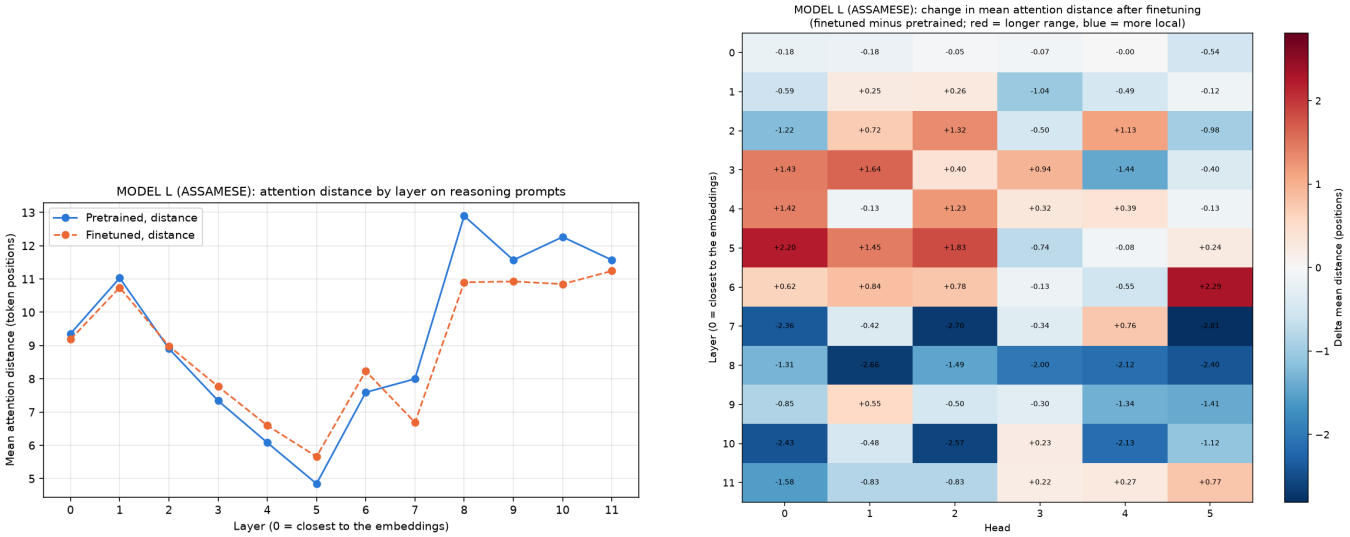


Figure 12: Model L attention distance before/after fine-tuning and per-head distance change. Negative distance delta indicates more local attention.

example, pretrained mean distance is about 12.9 token positions while the fine-tuned value is around 10.9.

The per-head delta shows both positive and negative changes. Several middle-layer heads increase their range, while layer 8 is strongly negative across most heads, with values such as −2.66, −2.00, −2.12 and −2.40. Layer 10 is also predominantly negative. This pattern is better described as a redistribution of attention range than as a global shift toward local or global attention.

4.10 3.10 Direct late-layer heatmap comparison

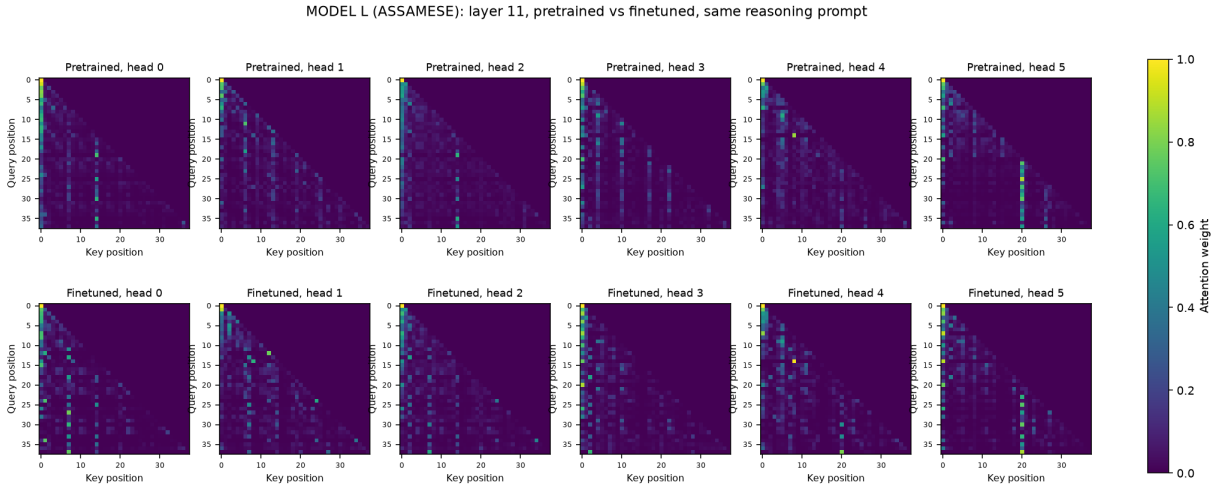


Figure 13: Model L, layer 11: pretrained versus fine-tuned attention heads for the same reasoning prompt.

The layer-11 heatmap gives a qualitative view of the numerical attention analysis. The final layer is close to the answer prediction, so changes there are particularly relevant when interpreting the increased attention to entity-name pieces. The figure should not be read as showing that the entire attention structure was replaced. Rather, the pretrained structure remains visible while selected attention patterns are altered by fine-tuning.

4.11 3.11 Language-model retention

The reasoning objective has a cost: the model must change enough to learn the new task, but excessive updates can destroy the pretrained language distribution. Run 3 provides the clearest warning. In the separate 600-window language-model retention evaluation used in Phase 3, perplexity increased from the pretrained baselines of 13.79 to 6709.43 for Telugu and from 61.94 to 868303.09 for Assamese. These pretrained baselines belong to the Phase 3 retention evaluation, not to the complete Phase 2 test-set evaluation reported in Table 1. The change is not ordinary overfitting in the sense of a small validation gap; the original language-model distribution was severely displaced.

Run 5 avoids this catastrophic degradation. On the separate 600-window language-model retention evaluation used in Phase 3, the final reported perplexities are 24.69 for Telugu and 129.19 for Assamese, compared with pretrained baselines of 13.79 and 61.94 respectively. These retention values should not be confused with the complete Phase 2 test-set results in Table 1, which report 32.7566 and 48.7110. The Phase 3 retention comparison is useful because the same smaller evaluation subset is reused across the fine-tuning checkpoints. On that retention evaluation, Run 5 therefore still increases perplexity relative to the corresponding pretrained checkpoint, so retention is not perfect; its value is that the increase is far smaller than under the aggressive configuration while useful reasoning performance is retained.

5 Integrated Discussion

The three phases explain different parts of the final behaviour. Phase 1 shows that the training distributions differ in source mixture and document structure. Phase 2 shows that the same architecture learns those distributions to different levels, with Model H reaching test perplexity 32.76 and Model L 48.71. Phase 3 then demonstrates that reasoning adaptation introduces another trade-off: improving the synthetic task can reduce the original language-model capability if the updates are too aggressive.

The tokenizer experiment is also connected to this capacity trade-off. Larger vocabularies give better compression, but embedding parameters grow linearly with vocabulary size. In a 25M-parameter model, moving from 8K to 32K would consume an additional 9.216M embedding parameters compared with 8K. The project therefore treats tokenizer selection as part of architecture design rather than a preprocessing choice made independently of the model budget.

The reasoning results show three distinct limitations. First, entity familiarity matters: held-out exact match is lower than seen-name exact match for both models. Second, reasoning depth matters: accuracy decreases consistently from one to three hops. Third, task type matters: direct numeric relation is much stronger than chain-extreme. These findings together suggest that the model learned useful local comparison behaviour but did not acquire a uniformly reliable multi-step reasoning procedure.

The internal analysis is consistent with this interpretation. Fine-tuning increases late-layer attention to entity-name pieces and reduces entropy in several middle-layer heads. At the same time, attention distance changes in both directions, indicating redistribution rather than a simple increase in attention range. This is compatible with a model that has become more selective about task-relevant information. It remains a correlation, however; an attention plot cannot establish the mechanism of reasoning by itself.

The five-run progression is perhaps the most practically useful part of the project. Run 1 showed that a small reasoning pool encouraged memorization without reliable name selection. Run 2 increased the pool and improved basic comparison behaviour, but the entity-selection problem remained. Run 3 demonstrated that an aggressive setup could force strong reasoning adaptation at an unacceptable language-model cost. Run 4 reduced that damage but did not recover enough reasoning ability. Run 5 changed the optimization process through low learning rate, warmup and curriculum sampling, producing the final measured balance.

6 Conclusion

The project demonstrates how a small multilingual language-model study can be analyzed at several levels instead of relying on a single final accuracy value. The corpus analysis explains the different training distributions, the tokenizer sweep shows the capacity trade-off between compression and model parameters, and pretraining establishes language-model baselines for both languages.

The reasoning phase adds a second objective and exposes the cost of adaptation. Fine-tuning improves controlled reasoning behaviour, but the improvement is weaker on held-out entities and decreases as the number of inference hops grows. The five-run progression shows why the final optimization configuration was changed: an aggressive setup produced severe language-model collapse, while a gentler curriculum-based setup provided a more controlled balance.

The attention measurements provide supporting evidence for structured internal change. Fine-tuning increases late-layer attention to entity-name pieces, sharpens several middle-layer heads and redistributes attention distance across layers and heads. These changes are consistent with greater selectivity for task-relevant information, but they do not by themselves prove a causal reasoning mechanism.

The main outcome of the project is therefore the combined experimental picture. Data composition affects the pretraining problem; pretraining establishes different language-model baselines; reasoning fine-tuning improves selected capabilities but exposes generalization and multi-hop limitations; and internal attention analysis shows that the adaptation is accompanied by non-uniform changes in how information is selected and propagated. The report deliberately keeps these claims within what the experiments actually demonstrate.